Exercise 4.2

SOLUTION

Q.1. Evaluate the following determinants.

i. $\begin{vmatrix} 1 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 3 & 2 \end{vmatrix}$

Sol: $\begin{vmatrix} 1 & -2 & -4 \\ 3 & -1 & -3 \\ -2 & 3 & 2 \end{vmatrix} = 1\begin{vmatrix} -1 & -3 \\ 3 & 2 \end{vmatrix} - (-2)\begin{vmatrix} 3 & -3 \\ -2 & 2 \end{vmatrix} + (-4)\begin{vmatrix} 3 & -1 \\ -2 & 3 \end{vmatrix}$ Expanding by 1st row

$= 1(-2+9) + 2(6-6) - 4(9-2)$

$= 1(7) + 2(0) - 4(7)$

$= 7 + 0 - 28 = -21$

ii. $\begin{vmatrix} a+b & a-b & a \\ a & a+b & a-b \\ a-b & a & a+b \end{vmatrix}$

Sol: $\begin{vmatrix} a+b & a-b & a \\ a & a+b & a-b \\ a-b & a & a+b \end{vmatrix} = (a+b)\begin{vmatrix} a+b & a-b \\ a & a+b \end{vmatrix} - (a-b)\begin{vmatrix} a & a-b \\ a-b & a+b \end{vmatrix} + a\begin{vmatrix} a & a+b \\ a-b & a \end{vmatrix}$

$= (a+b)\left[(a+b)^2 - a(a-b)\right] - (a-b)\left[a(a+b) - (a-b)^2\right] + a\left[a^2 - (a-b)(a+b)\right]$

$= (a+b)\left[a^2 + b^2 + 2ab - a^2 + ab\right] - (a-b)\left[a^2 + ab - a^2 - b^2 + 2ab\right] + a\left[a^2 - a^2 + b^2\right]$

$= (a+b)(b^2 + 3ab) - (a-b)(3ab - b^2) + a(b^2)$

$= ab^2 + 3a^2b + b^3 + 3ab^2 - 3a^2b + ab^2 + 3ab^2 - b^3 + ab^2 = 9ab^2$

iii. $\begin{vmatrix} 2x & x & x \\ y & 2y & y \\ z & z & 2z \end{vmatrix}$

Sol: $\begin{vmatrix} 2x & x & x \\ y & 2y & y \\ z & z & 2z \end{vmatrix}$

$= 2x\begin{vmatrix} 2y & y \\ z & 2z \end{vmatrix} - x\begin{vmatrix} y & y \\ z & 2z \end{vmatrix} + x\begin{vmatrix} y & 2y \\ z & z \end{vmatrix}$ Expending by 1st row

$= 2x(4yz - yz) - x(2yz - yz) + x(yz - 2yz)$

$= 2x(3yz) - x(yz) + x(-yz)$

$= 6xyz - xyz - xyz = 4xyz$

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Q.2: Without expansion show that:

i. $\begin{vmatrix} 7 & 8 & 9 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \end{vmatrix} = 0$

نوٹ:
ان سوالات میں اگر R.H.S برابر ہے zero کے تو آپ دیے گئے matrix کی rows یا columns کو ایک جیسا بنائیں یا کسی ایک Row یا column کے تمام elements کو zero بنائیں۔

Sol: L.H.S $= \begin{vmatrix} 7 & 8 & 9 \\ 5 & 6 & 7 \\ 2 & 3 & 4 \end{vmatrix}$

$C_2 - C_1$ and $C_3 - C_2$

$= \begin{vmatrix} 7 & 1 & 1 \\ 5 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = 0$ (Because C_2 and C_3 are identical)

ii. $\begin{vmatrix} 5 & 6 & -1 \\ 2 & 2 & 0 \\ 2 & -8 & 10 \end{vmatrix} = 0$

Sol: L.H.S $= \begin{vmatrix} 5 & 6 & -1 \\ 2 & 2 & 0 \\ 2 & -8 & 10 \end{vmatrix}$

Add C_2 in C_3

$= \begin{vmatrix} 5 & 6 & 5 \\ 2 & 2 & 2 \\ 2 & -8 & 2 \end{vmatrix} = 0 =$ R.H.S (Because C_1 in C_3 are identical)

iii. $\begin{vmatrix} -a & 0 & b \\ 0 & a & -c \\ c & -b & 0 \end{vmatrix} = 0$

Sol: L.H.S $= \begin{vmatrix} -a & 0 & b \\ 0 & a & -c \\ c & -b & 0 \end{vmatrix}$

$$= \frac{1}{abc} \begin{vmatrix} -ac & 0 & bc \\ 0 & ab & -bc \\ ac & -ab & 0 \end{vmatrix}, R_1 \times c,\ R_2 \times b,\ R_3 \times a$$

$$= \frac{1}{abc} \begin{vmatrix} -\cancel{ac} + \cancel{ac} & \cancel{ab} - \cancel{ab} & \cancel{bc} - \cancel{bc} \\ 0 & ab & -bc \\ ac & -ab & 0 \end{vmatrix} \text{Add } R_2, R_3 \text{ in } R_1$$

$$= \frac{1}{abc} \begin{vmatrix} 0 & 0 & 0 \\ 0 & ab & -bc \\ ac & -ab & 0 \end{vmatrix} = \frac{1}{abc}(0) = 0$$ (Because R_1 is zero)

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iv. $\begin{vmatrix} l & m+n & 1 \\ m & n+l & 1 \\ n & l+m & 1 \end{vmatrix} = 0$

Sol: L.H.S $= \begin{vmatrix} l & m+n & 1 \\ m & n+l & 1 \\ n & l+m & 1 \end{vmatrix}$

$= \begin{vmatrix} l+m+n & m+n & 1 \\ l+m+n & n+l & 1 \\ l+m+n & l+m & 1 \end{vmatrix}$ Add C_2 in C_1

$= (l+m+n)\begin{vmatrix} 1 & m+n & 1 \\ 1 & n+l & 1 \\ 1 & l+m & 1 \end{vmatrix}$ *Take* $(l+m+n)$ *common from* C_1

$= (l+m+n)(0) = 0$ (Because c_1 and c_3 are identical)

v. $\begin{vmatrix} 2 & 1 & 3x \\ 2 & 3 & 9x \\ 3 & 5 & 15x \end{vmatrix} = 0$

Sol: L.H.S $= \begin{vmatrix} 2 & 1 & 3x \\ 2 & 3 & 9x \\ 3 & 5 & 15x \end{vmatrix} = 3x\begin{vmatrix} 2 & 1 & 1 \\ 2 & 3 & 3 \\ 3 & 5 & 5 \end{vmatrix}$ take common $3x$ from C_3

$= 3x(0) = 0$ Because C_2 and C_3 are identical)

vi. $\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

Sol: L.H.S $= \begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$

$= \frac{1}{abc}\begin{vmatrix} abc & a^2 & a^3 \\ abc & b^2 & b^3 \\ abc & b^2 & b^3 \end{vmatrix} (a \times R_1, b \times R_2, c \times R_3)$

$= \frac{\cancel{abc}}{\cancel{abc}}\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$ Taking common abc from C_1

$= \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$ = R.H.S

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Q.3. Using properties of determinants, show that:

نوٹ:
اگر R.H.S یہ زیرو یا factor یا determinant ہوں تو operation سے حل کریں ورنہ صرف expand کر دیں۔

i. $\begin{vmatrix} 3 & 5 & 0 \\ 5 & 25 & 10 \\ 7 & 25 & 1 \end{vmatrix} = 25 \begin{vmatrix} 3 & 1 & 0 \\ 1 & 1 & 2 \\ 7 & 5 & 1 \end{vmatrix}$

Sol: L.H.S $= \begin{vmatrix} 3 & 5 & 0 \\ 5 & 25 & 10 \\ 7 & 25 & 1 \end{vmatrix} = 5 \begin{vmatrix} 3 & 5 & 0 \\ 1 & 5 & 2 \\ 7 & 25 & 1 \end{vmatrix}$ taking 5 common from R_2

$= 5.5 \begin{vmatrix} 3 & 1 & 0 \\ 1 & 1 & 2 \\ 7 & 5 & 1 \end{vmatrix}$ Taking 5 common from C_2

$= 25 \begin{vmatrix} 3 & 1 & 0 \\ 1 & 1 & 2 \\ 7 & 5 & 1 \end{vmatrix}$

= R.H.S

ii. $\begin{vmatrix} a+x & a & a \\ a & a+x & a \\ a & a & a+x \end{vmatrix} = x^2(3a+x)$

Sol: L.H.S $= \begin{vmatrix} a+x & a & a \\ a & a+x & a \\ a & a & a+x \end{vmatrix}$ Add R_2, R_3 in R_1

نوٹ:
R.H.S میں $3a+x$ سے اندازہ ہو جاتا ہے کہ R_2 اور R_3 کو R_1 میں جمع کر کے common لینا ہے۔

$= \begin{vmatrix} 3a+x & 3a+x & 3a+x \\ a & a+x & a \\ a & a & a+x \end{vmatrix}$

$= (3a+x) \begin{vmatrix} 1 & 1 & 1 \\ a & a+x & a \\ a & a & a+x \end{vmatrix}$ Take common $(3a+x)$ from R_1

$= (3a+x) \begin{vmatrix} 1 & 0 & 0 \\ a & x & 0 \\ a & 0 & x \end{vmatrix}$ $C_2 - C_1, C_3 - C_1$

By using property of determinant of a triangular matrix

$= (3a+x)(1.x.x)$

$= (3a+x)x^2$

$= x^2(3a+x) = R.H.S$

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iii. $\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$

Sol: $\text{L.H.S} = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$

$= \frac{1}{xyz} \begin{vmatrix} x & x^2 & xyz \\ y & y^2 & xyz \\ z & z & xyz \end{vmatrix}$ $x \times R_1, y \times R_2, z \times R_3$

$= \frac{\cancel{xyz}}{\cancel{xyz}} \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z & 1 \end{vmatrix}$ Take common xyz from C_3

$= \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = -\begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix}$ Interchange C_2 and C_3

$= (-)(-) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$ Again interchange C_2 and C_1

$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$ = R.H.S

iv. $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$

Sol: $\text{L.H.S} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$

$= \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$ $R_2 - R_1, R_3 - R_1$

$= \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & (y-x)(y+x) \\ 0 & z-x & (z-x)(z+x) \end{vmatrix}$

Take common $(y-x)$ from R_2 and $(z-x)$ from R_3

نوٹ:

اگر R.H.S پر factors بنے ہوں تو پہلے جس Column یا Row میں 1 ہے اسے باقی میں سے minus کر کے zero بنائیں۔ پھر جو دو factor ان میں سے Common آرہے ہوں common لیں پھر جس Column یا Row میں zero ہیں اسی سے open کریں۔

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$$=(y-x)(z-x)\begin{vmatrix}1 & x & x^2\\ 0 & 1 & y+x\\ 0 & 1 & z+x\end{vmatrix}$$

$$=(y-x)(z-x)\left|1\begin{vmatrix}1 & y+x\\ 1 & z+x\end{vmatrix}-0+0\right| \text{ Expand from } C_1$$

$$=(y-x)(z-x)\left(z+x-y-x\right)=(y-x)(z-x)(z-y)$$

$$=(-1)(x-y)(z-x)(-1)(y-z)=(-1)(-1)(x-y)(y-z)(z-x)=(x-y)(y-z)(z-x)=\text{R.H.S}$$

v. $$\begin{vmatrix}1 & 1 & 1\\ a+1 & b+1 & c+1\\ (a+1)^2 & (b+1)^2 & (c+1)^2\end{vmatrix}=(a-b)(b-c)(c-a)$$

Sol: $$\text{L.H.S}=\begin{vmatrix}1 & 1 & 1\\ a+1 & b+1 & c+1\\ (a+1)^2 & (b+1)^2 & (c+1)^2\end{vmatrix}$$

$$=\begin{vmatrix}1 & 1-1 & 1-1\\ a+1 & b+1-a-1 & c+1-a-1\\ (a+1)^2 & (b+1)^2-(a+1)^2 & (c+1)^2-(a+1)^2\end{vmatrix} \text{ by } C_2-C_1 \text{ and } C_3-C_1$$

$$=\begin{vmatrix}1 & 0 & 0\\ a+1 & b-a & c-a\\ (a+1)^2 & (b+1-a-1)(b+1+a+1) & (c+1-a-1)(c+1+a+1)\end{vmatrix}$$

$$=\begin{vmatrix}1 & 0 & 0\\ a+1 & b-a & c-a\\ (a+1)^2 & (b-a)(b+a+2) & (c-a)(c+a+2)\end{vmatrix}$$

$$=(b-a)(c-a)\begin{vmatrix}1 & 0 & 0\\ a+1 & b-a & c-a\\ (a+1)^2 & (b+a+2) & (c+a+2)\end{vmatrix} \text{ Take common } (b-a) \text{ from } C_2 \text{ and } (c-a) \text{ from } C_3$$

$$=(b-a)(c-a)\begin{vmatrix}1 & 0 & 0\\ a+1 & 1 & 1\\ (a+1)^2 & (b+a+2) & (c+a+2)\end{vmatrix} \text{ Expanding by } R_1$$

$$=(b-a)(c-a)1\left(1(c+a+2)-1(b+a+2)\right)=(b-a)(c-a)\left(c+a+2-b-a-2\right)$$

$$=(b-a)(c-a)(c-b)=(-1)(a-b)(-1)(b-c)(c-a)$$

$$=(a-b)(b-c)(c-a)=R.H.S$$

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vi. $\begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix} = 4a^2b^2c^2$

Sol: $L.H.S = \begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$

$= \begin{vmatrix} 2a^2+2b^2 & 2b^2+2c^2 & 2a^2+2c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ By $R_1+(R_2+R_3)$

$= 2\begin{vmatrix} a^2+b^2 & b^2+c^2 & a^2+c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$ 2Common from R_1

$= 2\begin{vmatrix} a^2+b^2 & b^2+c^2 & c^2+a^2 \\ -b^2 & 0 & -c^2 \\ -a^2 & -c^2 & 0 \end{vmatrix}$ By $\begin{matrix} R_2-R_1 \\ R_3-R_1 \end{matrix}$

$= 2\begin{vmatrix} 0 & b^2 & a^2 \\ -b^2 & 0 & c^2 \\ -a^2 & -c^2 & 0 \end{vmatrix}$ By $R_1+(R_2+R_3)$

Expand by R_1

$= 2\left[0-b^2\begin{vmatrix} -b^2 & -c^2 \\ -a^2 & 0 \end{vmatrix} + a^2\begin{vmatrix} -b^2 & 0 \\ -a^2 & -c^2 \end{vmatrix}\right]$

$= 2\left(-b^2\left(0-a^2c^2\right)+a^2\left(b^2c^2-0\right)\right)$

$= 2\left(a^2b^2c^2+a^2b^2c^2\right)$

$= 2\left(2a^2b^2c^2\right) = 4a^2b^2c^2 = R.H.S$

vii. $\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ a+b & b+c & c+a \end{vmatrix} = a^3+b^3+c^3-3abc$

Sol: $L.H.S = \begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ a+b & b+c & c+a \end{vmatrix}$

$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b+c & c+a & a+b \\ a+b & b+c & c+a \end{vmatrix}$ Add R_2 in R_1

$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ a+b & b+c & c+a \end{vmatrix}$ Take $(a+b+c)$ common from R_1

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$$= (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ b+c & a-b & a-c \\ a+b & c-a & c-b \end{vmatrix} \begin{matrix} \\ C_2 - C_1 \\ C_3 - C_1 \end{matrix}$$

$$= (a+b+c)\left[1\begin{vmatrix} a-b & a-c \\ c-a & c-b \end{vmatrix} - 0 + 0\right] \text{ Expand by } R_1$$

$$= (a+b+c)\left((a-b)(c-b)-(a-c)(c-a)\right)$$

$$= (a+b+c)\left[\left(\left(ac-ab-bc+b^2\right)-\left(ac-a^2-c^2+ac\right)\right)\right]$$

$$= (a+b+c)\left[\left(ac-ab-bc+b^2-ac+a^2+c^2-ac\right)\right] = (a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)$$

$$= a^3 + \cancel{ab^2} + \cancel{ac^2} - \cancel{a^2b} - abc - \cancel{ca^2} + \cancel{ba^2} + b^3 + \cancel{bc^2} - \cancel{ab^2} - \cancel{b^2c} - abc$$

$$\cancel{a^2c} + \cancel{cb^2} + c^3 - abc - \cancel{bc^2} - \cancel{c^2a} = a^3 + b^3 + c^3 - 3abc = R.H.S$$

viii. $\begin{vmatrix} a+t & a & a \\ b & b+t & b \\ c & c & c+t \end{vmatrix} = t^2(a+b+c+t)$

Sol: $L.H.S = \begin{vmatrix} a+t & a & a \\ b & b+t & b \\ c & c & c+t \end{vmatrix}$

$$= \begin{vmatrix} a+b+c+t & a+b+c+t & a+b+c+t \\ b & b+t & b \\ c & c & c+t \end{vmatrix} \text{ Add } R_2, R_3 \text{ in } R_1$$

$$= (a+b+c+t)\begin{vmatrix} 1 & 1 & 1 \\ b & b+t & b \\ c & c & c+t \end{vmatrix} \text{ Take common } (a+b+c+t) \text{ from } R_1$$

$$= (a+b+c+t)\begin{vmatrix} 1 & 0 & 0 \\ b & t & 0 \\ c & 0 & t \end{vmatrix} C_2 - C_1, C_3 - C_1$$

By Using property of determinant of a triangular Matrix

$$= (a+b+c+t)(1.t.t)$$

$$= t^2(a+b+c+t) = R.H.S$$

ix. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$

Sol: $L.H.S = \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

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$$= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \quad \text{By } R_1 + (R_2 + R_3)$$

$$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} \quad \text{By } (a+b+c) \text{ common } R_1$$

$$= (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ 2b & -a-b-c & 0 \\ 2c & 0 & -a-b-c \end{vmatrix} \quad \text{By } \begin{matrix} C_2 - C_1 \\ C_3 - C_1 \end{matrix}$$

Expand by R_1

$$= (a+b+c)\left[1(-a-b-c)(-a-b-c) - 0 + 0\right]$$

$$= (a+b+c)(-1)(a+b+c)(-1)(a+b+c)$$

$$= (a+b+c)^3 = R.H.S$$

x. $$\begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x+y+z)(x-y)(y-z)(z-x)$$

Sol: $$\text{L.H.S} = \begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \begin{vmatrix} x+y+z & x+y+z & x+y \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} \quad \text{Add } R_2 \text{ in } R_1$$

$$= (x+y+z)\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} \quad \text{Take common } (x+y+z) \text{ from } R_1$$

$$= (x+y+z)\begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \end{vmatrix} \quad C_2 - C_1 \text{ and } C_3 - C_1$$

$$= (x+y+z)\left|1\begin{vmatrix} y-x & z-x \\ y^2-x^2 & z^2-x^2 \end{vmatrix} - 0 + 0\right| \quad \text{Expand by } R_1$$

$$= (x+y+z)\begin{vmatrix} y-x & z-x \\ (y-x)(y+x) & (z-x)(z+x) \end{vmatrix}$$

$$= (x+y+z)(y-x)(z-x)\begin{vmatrix} 1 & 1 \\ (y+x) & z+x \end{vmatrix} \quad \text{Take common } (y-x) \text{ from } C_1 \text{ and } (z-x) \text{ from } C_2$$

$$= (x+y+z)(y-x)(z-x)\left(z+\cancel{x}-y-\cancel{x}\right) = (x+y+z)(y-x)(z-x)(z-y)$$

$$= (x+y+z)\left((-1)(x-y)\right)(z-x)(-1)(y-z) = (x+y+z)(x-y)(y-z)(z-x) = R.H.S$$

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xi. $\begin{vmatrix} 1 & 1 & 1 \\ a^2+1 & b^2+1 & c^2+1 \\ a^3+a & b^3+b & c^3+c \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca-1)$

Sol: L.H.S $= \begin{vmatrix} 1 & 0 & 0 \\ a^2+1 & b^2+1-a^2-1 & c^2+1-a^2-1 \\ a^3+a & b^3+b-a^3-a & c^3+c-a^3-a \end{vmatrix}$ by C_2-C_1 and C_3-C_1

$= \begin{vmatrix} 1 & 0 & 0 \\ a^2+1 & b^2-a^2 & c^2-a^2 \\ a^3+a & b^3-a^3+b-a & c^3-a^3+c-a \end{vmatrix}$

$= \begin{vmatrix} 1 & 0 & 0 \\ a^2+1 & (b-a)(b+a) & (b-a)(b+a) \\ a^3+a & (b-a)(b^2+ab+a^2)+(b-a) & (c-a)(c^2+ac+a^2)+(c-a) \end{vmatrix}$

$= (b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a^2+1 & (b+a) & c+a \\ a^3+a & b^2+ab+a^2+1 & c^2+ac+a^2+1 \end{vmatrix}$ Take $(b-a)$ common from C_2 and $(c-a)$ from C_3

$= (b-a)(c-a)\left[1\begin{vmatrix} b+a & c+a \\ b^2+ab+a^2+1 & c^2+ac+a^2+1 \end{vmatrix} - 0 + 0\right]$ Expand by R_1

$= (b-a)(c-a)\begin{vmatrix} b+a & c+a \\ b^2+ab+a^2+1 & c^2+ac+a^2+1 \end{vmatrix}$

$= (b-a)(c-a)\begin{vmatrix} b+a & c+a-b-a \\ b^2+ab+a^2+1 & c^2+ac+a^2+1-b^2-ab-a^2-1 \end{vmatrix} C_2-C_1$

$= (b-a)(c-a)\begin{vmatrix} b+a & c-b \\ b^2+ab+a^2+1 & c^2-b^2+ac-ab \end{vmatrix}$

$= (b-a)(c-a)\begin{vmatrix} b+a & c-b \\ b^2+ab+a^2+1 & (c-b)(c+b)+a(c-b) \end{vmatrix}$

$= (b-a)(c-a)(c-b)\begin{vmatrix} b+a & 1 \\ b^2+ab+a^2+1 & c+b+a \end{vmatrix}$ Take $(c-b)$ common from C_2

$= (-)(a-b)(-)(b-c)(c-a)\left[(b+a)(c+b+a)-b^2-ab-a^2-1\right]$

$= (a-b)(b-c)(c-a)\left[bc+b^2+ab+ac+ab+a^2-b^2-ab-a^2-1\right]$

$= (a-b)(b-c)(c-a)\left[ab+bc+ca-1\right] = R.H.S$

Hence Proved.

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xii. $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc+ab+bc+ca$

Sol: $L.H.S = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$

$= \begin{vmatrix} 1+a & 1 & 1 \\ -a & b & 0 \\ -a & 0 & c \end{vmatrix}$ By $\begin{matrix} R_2-R_1 \\ R_3-R_1 \end{matrix}$

$= \begin{vmatrix} 1+a & 1 & 0 \\ -a & b & -b \\ -a & 0 & c \end{vmatrix}$ By C_3-C_2

Expand by R_1

$= (1+a)(bc-0)-1(-ac-ab)-0$

$= bc+abc+ac+ab$

$= abc+ab+bc+ca = R.H.S$

xiii. $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} = 0$

Sol: $L.H.S = \begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$

$= \begin{vmatrix} 1 & a & a^2-bc \\ 0 & b-a & b^2-a^2+bc-ca \\ 0 & c-a & c^2-a^2+bc-ab \end{vmatrix}$ By R_2-R_1 and R_3-R_1

$= \begin{vmatrix} 1 & a & a^2-bc \\ 0 & (b-a) & (b-a)(b+a)+c(b-a) \\ 0 & (c-a) & (c-a)(c+a)+b(c-a) \end{vmatrix}$

$= (b-a)(c-a)\begin{vmatrix} 1 & a & a^2-bc \\ 0 & 1 & b+a+c \\ 0 & 1 & c+a+b \end{vmatrix}$ Take $(b-a)$ common from R_2 and $(c-a)$ from R_3

$\because R_2$ and R_3 are identical

$= (b-a)(c-a)(0) = 0 = R.H.S$

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xiv. $\begin{vmatrix} 2x+3 & x+2 & x+a \\ 2x+5 & x+3 & x+b \\ 2x+7 & x+4 & x+c \end{vmatrix} = 0$ *where* $2b = a+c$

Sol: $L.H.S = \begin{vmatrix} 2x+3 & x+2 & x+a \\ 2x+5 & x+3 & x+b \\ 2x+7 & x+4 & x+c \end{vmatrix}$

$= \begin{vmatrix} 2x+3 & x+2 & x+a \\ 2 & 1 & b-a \\ 4 & 2 & c-a \end{vmatrix}$ by $R_2 - R_1$ *and* $R_3 - R_1$

Given that $2b = a+c$ its mean a, b, c are in A.P so, $d = b-a$, $d = c-b$, $2d = c-a$

$= \begin{vmatrix} 2x+3 & x+2 & x+a \\ 2 & 1 & d \\ 4 & 2 & 2d \end{vmatrix}$

$= \begin{vmatrix} 2x+3 & x+2 & x+a \\ 2 & 1 & d \\ 2 & 1 & d \end{vmatrix}$ 2 common from R_3

$= 2(0)(R_2 \text{ and } R_3 \text{ are identical}) = 0 = R.H.S$

xv. $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$ where ω is an imaginary cube root of unity.

Sol: $L.H.S = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

$= \begin{vmatrix} a+b+c & b & c \\ a+b+c & a & b \\ a+b+c & c & a \end{vmatrix}$ By $C_1 + (C_2 + C_3)$

$= (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & a & b \\ 1 & c & a \end{vmatrix}$ Taking $(a+b+c)$ common from C_1

$= (a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & a-b & b-c \\ 0 & c-b & a-c \end{vmatrix}$ By $R_2 - R_1$ and $R_3 - R_1$

$= (a+b+c)\left\{\begin{vmatrix} a-b & b-c \\ c-b & a-c \end{vmatrix} - 0 + 0\right\}$ Expanding by C_1

$= (a+b+c)\{(a-b)(a-c) - (c-b)(b-c)\}$

$= (a+b+c)(a^2 - ac - ab + bc - bc + c^2 + b^2 - bc)$

$= (a+b+c)\{(a^2 + b^2 + c^2 - ab - bc - ac)\}$

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$$L.H.S = a^3 + b^3 + c^3 - 3abc$$

$$R.H.S = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$$

$$= (a+b+c)(a^2+ab\omega^2+ac\omega+ab\omega+b^2\omega^3+bc\omega^2+ac\omega^2+bc\omega^4+c^2\omega^3)$$

$$= (a+b+c)(a^2+b^2\omega^3+c^2\omega^3+ab(\omega^2+\omega)+bc(\omega^2+\omega^4)+ac(\omega+\omega^2))$$

Since ω is an imaginary cube root of unity, so

$$\omega+\omega^2 = -1, \omega^3 = 1, \omega^4 = \omega$$

$$R.H.S = (a+b+c)\{a^2+b^2(1)+c^2(1)+ab(-1)+bc(-1)+ac(-1)\}$$

$$= (a+b+c)\{a^2+b^2+c^2-ab-bc-ac\} = a^3+b^3+c^3-3abc$$

Hence proved $L.H.S = R.H.S$

Q.4. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -5 & 0 \\ -2 & -2 & 7 \end{bmatrix}$ *and* $B = \begin{bmatrix} -5 & -2 & 5 \\ -3 & -1 & 4 \\ -2 & -1 & 2 \end{bmatrix}$ then find

(i) A_{13}, A_{23}, A_{33} *and* $|A|$ (ii) B_{31}, B_{32}, B_{33} *and* $|B|$

Sol (i): Let $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -5 & 0 \\ -2 & -2 & 7 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 2 & -3 \\ 0 & -5 & 0 \\ -2 & -2 & 7 \end{vmatrix} = 1\begin{vmatrix} -5 & 0 \\ -2 & 7 \end{vmatrix} - 2\begin{vmatrix} 0 & 0 \\ -2 & 7 \end{vmatrix} + (-3)\begin{vmatrix} 0 & -5 \\ -2 & -2 \end{vmatrix} = 1(-35-0)-2(0-0)-3(0-10)$$

$$|A| = -35+30 \Rightarrow |A| = -5$$

$$A_{13} = (-1)^{1+3}\begin{vmatrix} 0 & -5 \\ -2 & -2 \end{vmatrix} = (-1)^4(0-10) = -10$$

$$A_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 2 \\ -2 & -2 \end{vmatrix} = (-1)^5(-2+4) = -(2) = -2$$

$$A_{33} = (-1)^{3+3}\begin{vmatrix} 1 & 2 \\ 0 & -5 \end{vmatrix} = (-1)^6(-5-0) = -5$$

ii. B_{31}, B_{32}, B_{33} *and* $|B|$

Sol: Let $B = \begin{bmatrix} -5 & -2 & 5 \\ -3 & -1 & 4 \\ -2 & -1 & 2 \end{bmatrix}$

$$|B| = \begin{vmatrix} -5 & -2 & 5 \\ -3 & -1 & 4 \\ -2 & -1 & 2 \end{vmatrix} = -5\begin{vmatrix} -1 & 4 \\ -1 & 2 \end{vmatrix} - (-2)\begin{vmatrix} -3 & 4 \\ -2 & 2 \end{vmatrix} + 5\begin{vmatrix} -3 & -1 \\ -2 & -1 \end{vmatrix} = -5(-2+4)+2(-6+8)+5(3-2)$$

$$= -5(2)+2(2)+5(1) = -10+4+5 = -10+9 = -1 \Rightarrow |B| = -1$$

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Now for B_{31}, B_{32}, B_{33}

$$B_{31}=(-1)^{3+1}\begin{vmatrix}-2 & 5\\ -1 & 4\end{vmatrix}=(-1)^4(-8+5)=1(-3)=-3$$

$$B_{32}=(-1)^{3+2}\begin{vmatrix}-5 & 5\\ -3 & 4\end{vmatrix}=(-1)^5(-20+15)=(-1)(-5)=5$$

$$B_{33}=(-1)^{3+3}\begin{vmatrix}-5 & -2\\ -3 & -1\end{vmatrix}=(-1)^6(5-6)=1(-1)=-1$$

Q.5. Find the values of x if

i. $\begin{vmatrix}2 & 1 & x\\ -1 & -4 & -3\\ x & 1 & 0\end{vmatrix}=5$ ii. $\begin{vmatrix}1 & x-1 & 3\\ -1 & x+1 & 2\\ 2 & -3 & x\end{vmatrix}=9$ iii. $\begin{vmatrix}1 & 1 & 1\\ 2 & x & 2\\ 3 & 6 & x\end{vmatrix}=0$

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i. $\begin{vmatrix}2 & 1 & x\\ -1 & -4 & -3\\ x & 1 & 0\end{vmatrix}=5$

Sol: Expand the determinant from R_1

$$2\begin{vmatrix}-4 & -3\\ 1 & 0\end{vmatrix}-1\begin{vmatrix}-1 & -3\\ x & 0\end{vmatrix}+x\begin{vmatrix}-1 & -4\\ x & 1\end{vmatrix}=5$$

$$2(0+3)-1(0+3x)+x(-1+4x)=5$$

$$6-3x-x+4x^2-5=0$$

$$4x^2-4x+1=0\Rightarrow(2x-1)^2=0$$

Taking square root on the both sides

$$(2x-1)=0\ \Rightarrow 2x=1\Rightarrow x=\frac{1}{2}$$

ii. $\begin{vmatrix}1 & x-1 & 3\\ -1 & x+1 & 2\\ 2 & -3 & x\end{vmatrix}=9$

Sol: $1\begin{vmatrix}x+1 & 2\\ -3 & x\end{vmatrix}-(x-1)\begin{vmatrix}-1 & 2\\ 2 & x\end{vmatrix}+3\begin{vmatrix}-1 & x+1\\ 2 & -3\end{vmatrix}=9$ Expand by R_1

$$(x(x+1)+6)-(x-1)(-x-4)+3(3-2(x+1))=9$$

$$(x^2+x+6)-(-x^2-4x+x+4)+3(3-2x-2)=9$$

$$(x^2+x+6)-(-x^2-3x+4)+3(-2x+1)=9$$

$$x^2+x+6+x^2+3x-4-6x+3=9$$

$$2x^2-2x+5=9\Rightarrow 2x^2-2x-4=0\Rightarrow x^2-x-2=0$$

$$x^2-2x+x-2=0\Rightarrow x(x-2)+1(x-2)=0$$

$$(x-2)(x+1)=0$$

$$\Rightarrow x-2=0 \quad or \quad x+1=0$$

$$x=2 \quad or \quad x=-1$$

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iii. $\begin{vmatrix} 1 & 1 & 1 \\ 2 & x & 2 \\ 3 & 6 & x \end{vmatrix} = 0$

Sol: $1\begin{vmatrix} x & 2 \\ 6 & x \end{vmatrix} - 1\begin{vmatrix} 2 & 2 \\ 3 & x \end{vmatrix} + 1\begin{vmatrix} 2 & x \\ 3 & 6 \end{vmatrix} = 0$

$1(x^2 - 12) - 1(2x - 6) + 1(12 - 3x) = 0$

$x^2 - 12 - 2x + 6 + 12 - 3x = 0$

$x^2 - 5x + 6 = 0$

$x^2 - 3x - 2x + 6 = 0$

$x(x-3) - 2(x-3) = 0$

$(x-2)(x-3) = 0$

$x - 2 = 0 \ or \ x - 3 = 0$

$x = 2 \ or \ x = 3$

Q.6. Find $|AA'|$ and $|A'A|$ if (*i*) $A = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$ (*ii*) $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix}$

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i. $A = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$

Sol: $A = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}$ then $A' = \begin{bmatrix} -3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix}$

$$AA' = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix}\begin{bmatrix} -3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 9+4+1 & -6+2-3 \\ -6+2-3 & 4+1+9 \end{bmatrix} = \begin{bmatrix} 14 & -7 \\ -7 & 14 \end{bmatrix}$$

$$|AA'| = \begin{vmatrix} 14 & -7 \\ -7 & 14 \end{vmatrix} = 196 - 49 = 147$$

Now, For $|A'A|$

$$A'A = \begin{bmatrix} -3 & 2 \\ 2 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} -3 & 2 & -1 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 9+4 & -6+2 & 3+6 \\ -6+2 & 4+1 & -2+3 \\ 3+6 & -2+3 & 1+9 \end{bmatrix} = \begin{bmatrix} 13 & -4 & 9 \\ -4 & 5 & 1 \\ 9 & 1 & 10 \end{bmatrix}$$

$$|A'A| = \begin{vmatrix} 13 & -4 & 9 \\ -4 & 5 & 1 \\ 9 & 1 & 10 \end{vmatrix}$$

$$|A'A| = 13\begin{vmatrix} 5 & 1 \\ 1 & 10 \end{vmatrix} - (-4)\begin{vmatrix} -4 & 1 \\ 9 & 10 \end{vmatrix} + 9\begin{vmatrix} -4 & 5 \\ 9 & 1 \end{vmatrix} = 13(50-1) + 4(-40-9) + 9(-4-45)$$

$$|A'A| = 13(49) + 4(-49) + 9(-49) = 637 - 196 - 441 = 0$$

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ii. $A=\begin{bmatrix}3 & 1\\ 2 & 2\\ 1 & 3\end{bmatrix}$

Sol: $A=\begin{bmatrix}3 & 1\\ 2 & 2\\ 1 & 3\end{bmatrix}$ then $A'=\begin{bmatrix}3 & 2 & 1\\ 1 & 2 & 3\end{bmatrix}$

$$AA'=\begin{bmatrix}3 & 1\\ 2 & 2\\ 1 & 3\end{bmatrix}\begin{bmatrix}3 & 2 & 1\\ 1 & 2 & 3\end{bmatrix}=\begin{bmatrix}9+1 & 6+2 & 3+3\\ 6+2 & 4+4 & 2+6\\ 3+3 & 2+6 & 1+9\end{bmatrix}$$

$$AA'=\begin{vmatrix}10 & 8 & 6\\ 8 & 8 & 8\\ 6 & 8 & 10\end{vmatrix} \text{ then } |AA'|=\begin{vmatrix}10 & 8 & 6\\ 8 & 8 & 8\\ 6 & 8 & 10\end{vmatrix}$$

$$|AA'|=10\begin{vmatrix}8 & 8\\ 8 & 10\end{vmatrix}-8\begin{vmatrix}8 & 8\\ 6 & 10\end{vmatrix}+6\begin{vmatrix}8 & 8\\ 6 & 8\end{vmatrix}$$

$$|AA'|=10(80-64)-8(80-48)+6(64-48)=10(16)-8(32)+6(16)=160-256+96$$

$$|AA'|=256-256=0$$

Now for $A'A$

$$A'A=\begin{bmatrix}3 & 2 & 1\\ 1 & 2 & 3\end{bmatrix}\begin{bmatrix}3 & 1\\ 2 & 2\\ 1 & 3\end{bmatrix}=\begin{bmatrix}9+4+1 & 3+4+3\\ 3+4+3 & 1+4+9\end{bmatrix}=\begin{bmatrix}14 & 10\\ 10 & 14\end{bmatrix} \text{ then}$$

$$|A'A|=\begin{vmatrix}14 & 10\\ 10 & 14\end{vmatrix}=14\times14-10\times10=196-100=96$$

Q.7. If A is a square matrix of order 3, then show that $|kA|=k^3|A|$

Sol: let $A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$ then $kA=k\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}=\begin{bmatrix}ka_{11} & ka_{12} & ka_{13}\\ ka_{21} & ka_{22} & ka_{23}\\ ka_{31} & ka_{32} & ka_{33}\end{bmatrix}$

$$|kA|=\begin{vmatrix}ka_{11} & ka_{12} & ka_{13}\\ ka_{21} & ka_{22} & ka_{23}\\ ka_{31} & ka_{32} & ka_{33}\end{vmatrix}$$ Take k common from R_1, R_2, R_3

$$=k.k.k\begin{vmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{vmatrix}$$

$=k^3|A|$ = R.H.S

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Q.8. Find the values of λ if A and B are singular & $A=\begin{bmatrix}4&2&3\\7&\lambda&6\\2&3&1\end{bmatrix}$, $B=\begin{bmatrix}-2&4&5\\1&-2&1\\2&\lambda&0\end{bmatrix}$

i. $A=\begin{bmatrix}4&2&3\\7&\lambda&6\\2&3&1\end{bmatrix}$

Sol: Given matrix is singular so $|A|=\begin{vmatrix}4&2&3\\7&\lambda&6\\2&3&1\end{vmatrix}=0$

$$4\begin{vmatrix}\lambda&6\\3&1\end{vmatrix}-2\begin{vmatrix}7&6\\2&1\end{vmatrix}+3\begin{vmatrix}7&\lambda\\2&3\end{vmatrix}=0$$

$$4(\lambda-18)-2(7-12)+3(21-2\lambda)=0$$

$$4\lambda-72-14+24+63-6\lambda=0$$

$$-2\lambda+1=0$$

$$-2\lambda=-1\Rightarrow\lambda=\frac{1}{2}$$

ii: $B=\begin{bmatrix}-2&4&5\\1&-2&1\\2&\lambda&0\end{bmatrix}$

Sol: Given matrix is singular so $|B|=0$

$$|B|=\begin{vmatrix}-2&4&5\\1&-2&1\\2&\lambda&0\end{vmatrix}=0$$

$$|B|=-2\begin{vmatrix}-2&1\\\lambda&0\end{vmatrix}-4\begin{vmatrix}1&1\\2&0\end{vmatrix}+5\begin{vmatrix}1&-2\\2&\lambda\end{vmatrix}=0$$

$$|B|=-2(0-\lambda)-4(0-2)+5(\lambda+4)=0$$

$$2\lambda+8+5\lambda+20=0$$

$$7\lambda+28=0\Rightarrow7\lambda=-28$$

$$\lambda=\frac{-28}{7}=-4\Rightarrow\lambda=-4$$

Q.9: Find the inverse of $A=\begin{bmatrix}1&2&1\\-5&0&4\\5&4&0\end{bmatrix}$ and show that $A^{-1}A=I_3$

Sol: $A=\begin{bmatrix}1&2&1\\-5&0&4\\5&4&0\end{bmatrix}$

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Then $|A| = \begin{vmatrix} 1 & 2 & 1 \\ -5 & 0 & 4 \\ 5 & 4 & 0 \end{vmatrix} = 1\begin{vmatrix} 0 & 4 \\ 4 & 0 \end{vmatrix} - 2\begin{vmatrix} -5 & 4 \\ 5 & 0 \end{vmatrix} + 1\begin{vmatrix} -5 & 0 \\ 4 & 4 \end{vmatrix}$

$= 1(0-16) - 2(0-20) + 1(-20-0) = -16 + 40 - 20 = 4$

$A_{11} = (-1)^{1+1}\begin{vmatrix} 0 & 4 \\ 4 & 0 \end{vmatrix} = (-1)^2(0-16) = -16$

$A_{12} = (-1)^{1+2}\begin{vmatrix} -5 & 4 \\ 5 & 0 \end{vmatrix} = (-1)^3(0-20) = -(-20) = 20$ $A_{13} = (-1)^{1+3}\begin{vmatrix} -5 & 0 \\ 5 & 4 \end{vmatrix} = (-1)^4(-20-0) = -20$

$A_{21} = (-1)^{2+1}\begin{vmatrix} 2 & 1 \\ 4 & 0 \end{vmatrix} = (-1)^3(0-4) = -(-4) = 4$

$A_{22} = (-1)^{2+2}\begin{vmatrix} 1 & 1 \\ 5 & 0 \end{vmatrix} = (-1)^4(0-5) = -5$

$A_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 2 \\ 5 & 4 \end{vmatrix} = (-1)^5(4-10) = -(-6) = 6$

$A_{31} = (-1)^{3+1}\begin{vmatrix} 2 & 1 \\ 0 & 4 \end{vmatrix} = (-1)^4(8-0) = 8$

$A_{32} = (-1)^{3+2}\begin{vmatrix} 1 & 1 \\ -5 & 4 \end{vmatrix} = (-1)^5(4+5) = -(9) = -9$

$A_{33} = (-1)^{3+3}\begin{vmatrix} 1 & 2 \\ -5 & 0 \end{vmatrix} = (-1)^6(0+10) = (+1)(10) = +10$

Cofactor of $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} = \begin{bmatrix} -16 & 20 & -20 \\ 4 & -5 & 6 \\ 8 & -9 & +10 \end{bmatrix}$

$AdjA = (cofactorA)' = \begin{bmatrix} -16 & 4 & 8 \\ 20 & -5 & -9 \\ -20 & 6 & +10 \end{bmatrix}$

$A^{-1} = \dfrac{Adj}{|A|} = \dfrac{1}{4}\begin{bmatrix} -16 & 4 & 8 \\ 20 & -5 & -9 \\ -20 & 6 & +10 \end{bmatrix}$

$A^{-1}A = \dfrac{1}{4}\begin{bmatrix} -16 & 4 & 8 \\ 20 & -5 & -9 \\ -20 & 6 & +10 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ -5 & 0 & 4 \\ 5 & 4 & 0 \end{bmatrix} = \dfrac{1}{4}\begin{bmatrix} -16-20+40 & -32+0+32 & -16+16+0 \\ 20+25-45 & 40+0-36 & 20-20+0 \\ -20-30+50 & -40+0+40 & -20+24+0 \end{bmatrix}$

$A^{-1}A = \dfrac{1}{4}\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \dfrac{\cancel{4}}{\cancel{4}}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$

Hence proved $A^{-1}A = I_3$

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Q.10: Verify that $(AB)' = B'A'$ If

(i) $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ -3 & -2 \\ 0 & 1 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 2 & 1 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}$

Sol: $AB = \begin{bmatrix} 1 & -1 & 2 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -3 & -2 \\ 0 & 1 \end{bmatrix}$

$$AB = \begin{bmatrix} 1+3+0 & 1+2+2 \\ 0+9+0 & 0+6+1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 9 & 7 \end{bmatrix}$$

$$\text{L.H.S} = (AB)' = \begin{bmatrix} 4 & 9 \\ 5 & 7 \end{bmatrix}$$

Now $B' = \begin{bmatrix} 1 & -3 & 0 \\ 1 & -2 & 1 \end{bmatrix}, A' = \begin{bmatrix} 1 & 0 \\ -1 & -3 \\ 2 & 1 \end{bmatrix}$

$$\text{R.H.S} = B'A' = \begin{bmatrix} 1 & -3 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & -3 \\ 2 & 1 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 1+3+0 & 0+9+0 \\ 1+2+2 & 0+6+1 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 5 & 7 \end{bmatrix}$$

Hence prove that $(AB)' = B'A'$

L.H.S = R.H.S

ii: $A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 2 & 1 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}$

Sol: $AB = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}$

$$AB = \begin{bmatrix} 1-4 & -3+2 \\ 1-8 & -3+4 \\ 2-2 & -6+1 \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ -7 & 1 \\ 0 & -5 \end{bmatrix}$$

$$AB = \begin{bmatrix} -3 & -1 \\ -7 & 1 \\ 0 & -5 \end{bmatrix}$$

$$\text{L.H.S} = (AB)' = \begin{bmatrix} -3 & -7 & 0 \\ -1 & 1 & -5 \end{bmatrix}$$

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Now

$$B' = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}, \quad A' = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 1 \end{bmatrix}$$

R.H.S = $B'A'$

$$B'A' = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 1 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 1-4 & 1-8 & 2-2 \\ -3+2 & -3+4 & -6+1 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} -3 & -7 & 0 \\ -1 & 1 & -5 \end{bmatrix}$$

Hence Prove that $(AB)' = B'A' \Rightarrow$ L.H.S = R.H.S

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